

- (d) the atomic number
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2022

Ans. (b)

See the explanation of above question.

37. **Isotopes are those atoms of the same element which have–**
(a) Atomic mass is same but atomic number is different
(b) Atomic mass is different but atomic number is same
(c) Atomic number and atoms mass both are same
(d) None of the above.

U.P. U.D.A./L.D.A. (Pre) 2001

Ans. (b)

See the explanation of above question.

38. **Isotopes is that nuclei of atomic nucleus in which –**
(a) Number of neutrons is same but number of protons is different
(b) Number of protons is same but number of neutrons is different
(c) Number of both protons and neutrons is same
(d) Number of both protons and neutrons is different

41st B.P.S.C. (Pre) 1996

Ans. (b)

See the explanation of above question.

39. **The isoneutronic group, in the following, is :**

- (a) ${}^6\text{C}_{12}$, ${}^7\text{N}_{14}$, ${}^8\text{O}_{16}$ (b) ${}^6\text{C}_{12}$, ${}^7\text{N}_{14}$, ${}^8\text{N}_{18}$
(c) ${}^6\text{C}_{14}$, ${}^7\text{N}_{14}$, ${}^8\text{O}_{16}$ (d) ${}^6\text{C}_{14}$, ${}^7\text{N}_{15}$, ${}^8\text{O}_{16}$

R.A.S./R.T.S. (Pre) 1999

Ans. (d)

Two nuclides are isoneutronic (isotones) if they have the very same neutron number N, but different proton number Z. For example, Boron-12 and Carbon-13 nuclei both contain 7 neutrons, and so are Isotones.

As per the question

$${}^6\text{C}_{14} = 14 - 6 = 8 \text{ neutron}$$

$${}^7\text{N}_{15} = 15 - 7 = 8 \text{ neutron}$$

$${}^8\text{O}_{16} = 16 - 8 = 8 \text{ neutron}$$

Since the number of neutrons in all is 8 they are all isotones.

40. **Radioactivity is measured by –**

- (a) Hydrometer (b) Geiger Counter
(c) Seismometer (d) Ammeter

U.P.P.S.C. (GIC) 2010

Ans. (b)

The Geiger–Muller Counter also called a Geiger Counter, is an instrument used for detection and measurement of ionizing radiation. It detects radiation such as alpha particles, beta particles and gamma rays using the ionization produced in a Geiger-Muller tube, which gives its name to the instrument.

41. **Radioactivity was discovered by :**

- (a) Rutherford (b) Becquerel
(c) Bohr (d) Madam Curie

U.P.P.C.S. (Mains) 2003

Ans. (b)

Radioactivity was discovered in 1896 by the French Scientist Henri Becquerel when he was working with phosphorescent materials.

Physical and Chemical Changes, Solution

Notes

- Matter is defined as anything that occupies space and exhibits rest mass or inertia.
- The three commonly occurring states of matter include solid, liquid and gas.
- In our daily life, we encounter many changes in our surroundings, though we do not notice them, such as souring of milk, rusting of iron, stretching of a rubber band, bread becoming toast, melting of wax, etc.
- All these changes that take place around us are either a physical change or chemical change.

Physical Changes

- Physical changes, are the changes that change the physical traits of the substance, without making any change in their internal structure.
- In physical changes no chemical bonds are broken or formed.
- Some examples of physical changes are :
(i) Melting of wax
(ii) Melting of ice
(iii) Freezing of water
(iv) Vaporization of water
(v) Dissolving sugar in water
(vi) Chopping in wood
(vii) Crumpling of paper
- Physical change is a reversible process.

Chemical Changes

- A chemical change is defined as the process in which the atoms of one or more substances are rearranged or combine to form a new substance.
- When a substance undergoes a chemical change, the chemical properties of the substance changes and it is

transformed into a different substance with different chemical composition.

- Energy change is one of the characteristics of a chemical change, because of the formation of a new product.
- Once the chemical change takes place, it cannot be reversed.
- Some examples of chemical changes are as follows -
 - (i) Adding vinegar to baking soda
 - (ii) Bleaching a stain
 - (iii) Fermenting of grapes
 - (iv) Burning of magnesium wire and paper
 - (v) Nail rusting
 - (vi) Sour of milk
 - (vii) Cooking of food

Comparison chart of Physical and Chemical changes

Basis for Comparison	Physical Change	Chemical Change
Meaning	Physical change refers to a change in which the molecules are rearranged but their internal composition remains same.	Chemical change is a process in which the substance transforms into a new substance, having different chemical composition.
Nature	Reversible	Irreversible
Original Matter	Can be recovered	Cannot be recovered
Involves	Change in physical properties of the substance, i.e. shape, size, colour etc.	Change in physical and chemical properties and composition of the substance.
Product Formation	No new product is formed.	New product is formed.
Energy	Very little energy (or no energy) is either absorbed or evolved.	Absorption and evolution of energy take place, during reaction.

Solution

- In chemistry, a solution is a special type of homogeneous mixture composed of two or more substances.
- Other characteristics of the solutions are -
 - (i) The particles of solute in a solution cannot be seen by the naked eye.

- (ii) A solution does not allow beams of light to scatter.
- (iii) A solution is stable.
- (iv) From a solution, the solute cannot be separated by filtration.

Solvent :

- Solvents are molecules that have the ability to dissolve other molecules, known as solutes.
- A solvent can be solid, liquid or gas.
- The molecules of the solvent work to put the solute molecules apart.
- Eventually, the molecules of solute become eventually distributed in throughout the solvent.

Types of Solution :

1. Gaseous Solution :

- If the solvent is gas, only gases are dissolved under a given set of conditions.
- Example of a gaseous solution is air (Oxygen and other gases dissolved in Nitrogen).

2. Liquid Solution :

- If the solvent is a liquid, then almost all gases, liquids and solids can be dissolved.
- Examples of liquid solutions are as follows-

(i) **Gas in liquid** – Oxygen in water.

(ii) **Liquid in liquid** – Alcoholic beverages are basically solutions of ethanol in water.

(iii) **Solid in liquid** – Sucrose in water; Sodium chloride in water.

3. Solid solutions :

- If the solvent is solid, then gas, liquid and solid can be dissolved.

(i) **Gas in Solid** – Hydrogen dissolves in palladium, this is studied as a means of hydrogen storage.

(ii) **Liquid in solid** – Mercury in gold forming amalgam; Hexane in paraffin wax.

(iii) **Solid in solid** – Alloys like bronze is a solution of copper and zinc.

Dilute Solution :

- A dilute solution in chemistry is a solution that can accept more solute, until it becomes saturated, or in which there is a relatively small amount of solute dissolved in the solution.

Concentrated Solution :

- The concentrated solution is a solution that contains a large amount of solute relative to the amount that could dissolve.

Saturated solution :

- A saturated solution is a solution that can not dissolve any more of the substance that's been mixed into it.

Solubility :

- Amount of a substance (called the solute) that dissolves in a unit volume of a liquid substance (called the solvent) to form a saturated solution under specified conditions of temperature and pressure.
- Solubility is expressed usually as moles of solute per 100 grams of solvent.

Colloids :

- A colloid is a mixture in which one substance consisting of microscopically dispersed insoluble particles is suspended throughout another substance. Colloids are classified according to the state of the dispersed phase and the dispersion medium.

Name of Colloid	Dispersed Phase	Dispersion Medium	Examples
Aerosol	Solid	Gas	Smoke, haze, dust
Liquid Aerosol	Liquid	Gas	Cloud, mist, fog
Emulsion	Liquid	Liquid	Milk, cream, mayonnaise
Gel	Liquid	Solid	Jelly, butter, curd, cheese
Sol	Solid	Liquid	Gold sol, sulphur sol
Solid-sol	Solid	Solid	Coloured glasses, gems, ruby glass
Foam	Gas	Liquid	Whipped cream, soap Lather
Solid foam	Gas	Solid	Pumice stone, rubber

Osmosis :

- Movement of solvent (such as water) through a semi-permeable membrane into a solution of higher solute concentration that tends to equalize the concentration of solute on the two sides of the membrane.

Semipermeable Membrane :

- A semipermeable membrane is any biological or synthetic membrane that allows certain molecules or ions to pass through while restricting others.
- A semipermeable membrane has spaces between the solid portions of its structure that enables certain substances or molecules to pass through but are designed to restrict others that are of larger size.

Question Bank**1. An example of physical change –**

- Blackening of silverware
- Burning of Candle
- Making of curd from milk
- Dissolving of sugar in water

38th B.P.S.C. (Pre) 1992**Ans. (d)**

Physical changes affect the form of a chemical substance, but not its chemical composition. This fact contrasts with the concept of chemical change in which the chemical composition of the substance changes or one or more substances combines or break up to form new substances. In general, a physical change is using physical means. For example, sugar dissolved in water can be recovered by allowing the water to evaporate.

2. Change of water into the vapour is called –

- Natural
- Physical
- Chemical
- Biological

U.P.P.C.S. (Pre) 1992**Ans. (b)**

Change of water into the vapour is a physical change.

3. Which of the following is an example of chemical change–

- Dispersion of white light into the light of 7 colours when passed through a prism
- Softening of vegetables when cooked
- Wet soil lump, when dried, gets fragile
- Dissolving of salt in water

R.A.S./R.T.S. (Pre) 2003**Ans. (b)**

A change which alters the specific properties of a material by bringing about a change in its molecular composition, followed by a change in state, is called a chemical change. The chemical change is a change in which something new is formed. Cooking of vegetables is a chemical change as it brings irreversible changes in the composition of vegetables.

4. Which of the following is/are the example/examples of chemical change?

- Crystallization of sodium chloride
- Melting of ice
- Souring of milk

Select the correct answer using the code given below.

- 1 and 2 only
- 3 only

- (c) 1, 2 and 3 (d) None

I.A.S. (Pre) 2014

Ans. (b)

A chemical change is a change in which something new is formed (irreversible). The starting materials change into an entirely different substance or substances. This new substance has a different chemical composition than the starting materials. Examples of chemical change would be the reaction of iron with air (rusting) or the reaction of a metal and acid. The crystallization of sodium chloride and melting of ice, both are the examples of a physical change while souring of milk is an example of chemical change.

5. Detect the odd one :

- (a) Ripening of fruit
(b) Setting of cement
(c) Burning of coal
(d) More than one of the above
(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (e)

Ripening of fruits, setting of cement and burning of coal—all of these are chemical changes and all are also exothermic. So there is no difference. Hence, option (e) is the correct answer.

6. Assertion (A) : A chemical reaction becomes faster at a higher temperature.

Reason (R) : At higher temperature, molecular motion becomes more rapid.

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true but (R) is not a correct explanation of (A).
(c) (A) is true but (R) is false.
(d) (A) is false but (R) is true.

I.A.S. (Pre) 2001

Ans. (a)

A chemical reaction generally becomes faster at a higher temperature because at higher temperature, molecular motion becomes more rapid and the molecules have higher kinetic energy. The process of a chemical reaction depends on the concentration, temperature and physical state of molecules and their nature. Normally, higher the temperature of molecules, faster would be the reaction. Hence, both (A) and (R) are true and (R) is the correct explanation of (A).

7. What form of energy is produced in the decomposition of water?

- (a) Light (b) Heat
(c) Sound (d) Acid

M.P.P.C.S. (Pre) 1997

Ans. (b)

Hydrolysis is the chemical reaction in which ions of a salt dissolved in water, mutually combines with a water molecule to form an acidic or basic solution and the energy produced in this process is in the form of heat.

8. Pasteurization is a process by which –

- (a) Milk is preserved at a very low temperature for 24 hours
(b) Milk is heated for 8 hours
(c) First milk is heated up to a longer time and then cooled suddenly within specified time
(d) None of these

38th B.P.S.C. (Pre) 1992

Ans. (c)

In order to pasteurize milk, first, the milk is heated up to a longer time and then cooled suddenly within specified time. Pasteurization is a process invented by French scientist Louis Pasteur during the nineteenth century. In 1864, Louis Pasteur discovered that heating beer and wine was enough to kill most of the bacteria that caused spoilage and hence, prevented these beverages from turning sour. This was achieved by eliminating pathogenic microbes and lowering microbial numbers to prolong the quality of the beverage.

9. Pasteurization is related to :

- (a) Sterilization of milk (b) Dehydration of milk
(c) Fermentation of milk (d) Distillation of milk

M.P. PCS (Pre) 2016

Ans. (a)

The Pasteurization is a process that kills microbes (mainly bacteria) in food and drinks such as milk and facilitates its preservation and safety. In this process, the milk is heated at about 63° C for 30 minutes or, alternatively, at about 72° C for 15 seconds to ensure that any harmful bacteria is not alive. The process was invented by the French scientist Louis Pasteur. So it is called Pasteurization.

10. Which of the following is NOT correctly matched?

- (a) Desalination Sea water
(b) Reverse osmosis Potable water
(c) Denaturation Proteins
(d) Pasteurization Tea

U.P.P.C.S. (Pre) 2019

Ans. (d)

Pasteurization is the partial sterilization of a product, such as milk, wine or fruit juices to make it safe for consumption and improve its keeping quality. Pasteurization was developed by Louis Pasteur in 1864. It is not used for tea. Other pairs are correctly matched.

11. **Phosphatase test is used for the analysis of :**

- (a) Milk
- (b) Tea
- (c) Water
- (d) All of the above

M.P. P.C.S. (Pre) 2021

Ans. (d)

Phosphatase test is commonly used for the analysis of milk and dairy products. The alkaline phosphatase (ALP) is an enzyme normally present in raw milk and it is inactivated in conditions of heat treatment. The temperature of inactivation of ALP is slightly higher than that required for the destruction of pathogenic bacteria. So the ALP test in pasteurized milk is used to verify if the heating process of pasteurization is done correctly. Phosphatase test is also used for checking the pollution of water. Acid phosphatase test proves superior to standard phenotypic identification procedure for *Clostridium perfringens* (It is used as an indicator for persistent faecal pollution as well as to monitor the efficacy of water treatment processes) strains isolated from water. ALP test may also be used for checking the quality of milk tea. It may be the reason that MPPSC has given answer (d) for this question in its final answer key.

An ALP blood test is also used for screening, diagnosis, and monitoring of liver or bone diseases and other health conditions. Alkaline phosphatase (ALP) enzyme is present in many parts of the body, but it is primarily found in the liver, bones, intestine, and kidneys. ALP testing measures the amount of this enzyme in the blood. Abnormal levels of ALP can be caused by liver or bone conditions and other types of health problems.

12. **Because of impurities the boiling point of a liquid –**

- (a) Increases
- (b) Decreases
- (c) Remains constant
- (d) They are not related

M.P.P.C.S. (Pre) 2004

Ans. (a)

The boiling point of a liquid is the temperature at which its vapour pressure is equal to the pressure of the gas above it. When some impurities are mixed with any liquid, its boiling point increases.

13. **Give a thought to these statements :**

Statement (A) : Temperature below 0°C is achieved by the mixture of salt and ice.

Reason (R) : Salt increases the freezing point of ice. In the context of above statements which one of the following is correct:

- (a) (A) and (R) both correct, and (R) is the correct explanation of (A).
- (b) (A) and (R) both correct, but (R) is not the correct explanation of (A).

(c) (A) is correct but (R) is wrong.

(d) (A) is wrong but (R) is correct.

U.P. Lower Sub. (Pre) 1998

I.A.S. (Pre) 1993

Ans. (c)

When some impurities are mixed with any liquid, its boiling point increases. On the other hand, the freezing point generally gets lowered below 0°C when ice is mixed with salts (NaCl or KNO₃ etc). Its mixture is called freezing-mixture. That is why statement (A) is correct, while reason (R) is wrong.

14. **The amount of solute present per unit volume or per unit mass of the solution/solvent is known as ____ in Chemistry.**

- (a) composition of solute
- (b) concentration of solvent
- (c) concentration of solute
- (d) concentration of solution

69th B.P.S.C. (Pre) 2023

Ans. (d)

The amount of solute present per unit volume or per unit mass of the solution/solvent is called 'Concentration of Solution'. The concentration of a solution is the measure of how much solute is present in a fixed amount (mass or volume) of the solution or the amount of solute dissolved in a given mass or volume of solvent.

15. **For two liquids A and B to form an ideal solution :**

- (a) Entropy of mixing must be zero
- (b) Free energy of mixing should be zero
- (c) Enthalpy of mixing should be zero
- (d) None of the above

70th B.P.S.C. (Pre) (Re-Exam) 2024

Ans. (c)

An ideal solution is a homogeneous mixture where the interactions between solute and solvent molecules are identical to those between the pure components. This means that the intermolecular forces between solute and solvent molecules are roughly the same as the intermolecular forces between solute-solute and solvent-solvent molecules. For an ideal solution, the enthalpy of mixing (ΔH_{mix}) is zero. The enthalpy of mixing (ΔH_{mix}) is the change in enthalpy that occurs when a substance is mixed with another substance or when a solution is formed. In an ideal solution, since the interactions between molecules are similar, there is no net change in energy when the components are mixed. Therefore, the enthalpy of mixing is zero. This means that when the

components of an ideal solution are mixed, there is no heat absorbed or released. In essence, the molecules mix without any energy change. The ideal solution follows Raoult's Law, which states that the partial vapour pressure of each component in the solution is directly proportional to its mole fraction.

16. Consider the following statements :

- 1. Addition of salt to pure water increases the boiling point of water and decreases the freezing point of water.**
- 2. When methyl alcohol is added to water, boiling point of water decreases.**

Which of the above statements is/are correct?

- (a) Salt and methyl alcohol are same in behaviour
- (b) 1 only
- (c) 2 only
- (d) Both 1 and 2
- (e) None of the above/More than one of the above

67th B.P.S.C. (Pre) (Re. Exam) 2022

Ans. (d)

When salt is dissolved in pure water, it increases the boiling point and decreases the freezing point of water. While when methyl alcohol is added to water, it decreases the boiling point of water. This happens due to hydrogen bonding.

17. The conductivity of a strong electrolyte :

- (a) Decreases on dilution
- (b) Does not changes with dilution
- (c) Increases slightly on dilution
- (d) None of the above

70th B.P.S.C. (Pre) (Re-Exam) 2024

Ans. (a)

The conductivity of a strong electrolyte generally decreases slightly on dilution. Conductivity of an electrolyte solution is directly related to the concentration and mobility of ions in the solution. Strong electrolytes, like strong acids, strong bases, and soluble ionic salts, completely dissociate into ions when dissolved. When a strong electrolyte solution is diluted, the volume of the solution increases while the number of ions remains relatively constant. This leads to a decrease in the concentration of ions per unit volume. While dilution increases the distance between ions and slightly improves their mobility, the decrease in ion concentration outweighs this effect in strong electrolytes, resulting in a net decrease in conductivity. Examples of strong electrolytes are NaCl, KCl, etc.

18. Milk is an example of :

- (a) A gel
- (b) A foam
- (c) An emulsion
- (d) A suspension

42nd B.P.S.C. (Pre) 1997

U.P.P.C.S. (Pre) 1996

Ans. (c)

An emulsion is a colloid of two or more immiscible liquids where one liquid contains a dispersion of the other liquids. Milk is an example of an emulsion. It is an example of an oil in water emulsion.

A colloid is a mixture in which one substance consisting of microscopically dispersed insoluble particles is suspended throughout another substance. Colloids are classified according to the state of the dispersed phase and the dispersion medium.

Name of Colloid	Dispersed Phase	Dispersion Medium	Examples
Aerosol	Solid	Gas	Smoke, haze, dust
Liquid Aerosol	Liquid	Gas	Cloud, mist, fog
Emulsion	Liquid	Liquid	Milk, cream, mayonnaise
Gel	Liquid	Solid	Jelly, butter, curd, cheese
Sol	Solid	Liquid	Gold sol, sulphur sol
Solid-sol	Solid	Solid	Coloured glasses, gems, ruby glass
Foam	Gas	Liquid	Whipped cream, soap Lather
Solid foam	Gas	Solid	Pumice stone, rubber

19. Milk is an example of :

- (a) Foam
- (b) Sol
- (c) Aerosol
- (d) Emulsion

U.P.P.C.S. (Pre) 2024

Ans. (d)

See the explanation of above question.

20. Which of these is not a colloid :

- (a) Milk
- (b) Blood
- (c) Icecream
- (d) Honey

U.P.P.C.S. (Pre) 2002

Ans. (d)

Among the given options, honey is not a colloid. At room temperature, honey is a supercooled liquid, in which the glucose precipitates into solid granules. This forms a semisolid solution of precipitated glucose crystals in a solution of fructose and other ingredients. Milk, blood and icecream are colloids.

21. Which type of colloidal system is expressed in the fog :

- (a) Liquid in a gas (b) Gas in a liquid
(c) Solid in a gas (d) Liquid in Liquid

I.A.S. (Pre) 1993

Ans. (a)

When the water vapour in the air condensed to form small droplets of water, the fog is formed. When the temperature of air during winter season reaches near to dew point, the water vapour gets condensed around small particles of dust and remains suspended in the air because of their low weight.

22. Given below are two statements, one is labelled as Assertion (A) and the other as Reason (R).

Assertion (A) : A solution of table salt in a glass of water is homogeneous.

Reason (R) : A solution having different compositions throughout is homogeneous.

Select the correct answer from the code given below :

Code :

- (a) (A) is true but (R) is false
(b) Both (A) and (R) are true and (R) is the correct explanation of (A)
(c) (A) is false but (R) is true
(d) Both (A) and (R) are true but (R) is not the correct explanation of (A)

U.P. P.C.S. (Pre) 2022

Ans. (a)

Homogeneous solutions have the same composition and properties throughout the solution. None of the components involved in the homogeneous solution can be detected or observed separately. A solution of table salt in a glass of water is an example of a homogeneous solution. A solution having different compositions and properties throughout is a heterogeneous mixture. Individual components of a heterogeneous mixture can be discreetly observed. Hence, option (a) is the correct answer.

23. If an unsaturated sugar solution is prepared by mixing 100 grams of sugar with half litre of water, which of the following physical quantities will not change?

- (a) Mass (b) Density

(c) Concentration

(d) Volume

U.P.P.C.S. (Pre) 2017

Ans. (d)

If an unsaturated sugar solution is prepared by mixing 100 grams of sugar with half liter of water then volume of the solution remain constant.

24. Water passes from the soil into the roots by a physical process called :

- (a) diffusion (b) transpiration
(c) absorption (d) osmosis
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2022

Ans. (d)

The plants absorb water from the soil with the help of root hairs through osmosis. The diffusion of water molecules through a semi-permeable membrane is called osmosis. The water moves into root hair cells by osmosis, as there is a high concentration of minerals/sugars in the cells, and a little water. This decreases the water potential inside the cell, so water molecules move down gradient into the cells. The water can then be moved by xylem in the transpiration stream through the plant and out of its leaves.

25. After cutting cucumber if salt is added, then water comes out due to :

- (a) Active transport (b) Passive transport
(c) Osmosis (d) Diffusion

U.P. P.C.S. (Pre) 2021

Ans. (c)

Osmosis is the process when water (or solvent) molecules move from a region (solution) of low concentration to a region(solution) of higher concentration through a semi-permeable membrane. When salt is applied to cucumber or cut onion pieces, it makes their surrounding more concentrated than the concentration of their cell sap. Then it causes the water molecules in the cells of the cucumber and the cut onion pieces to move to the surrounding through their cell membrane (which is semipermeable).

26. Consider the following statements :

- 1. During the process of osmosis, the solvent travels from the concentrated solution to the dilute solution.**
- 2. In the reverse osmosis, external pressure is applied to the dilute solution.**

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

I.A.S. (Pre) 2005

Ans. (d)

In the process of osmosis, the solvent moves through a semi-permeable membrane into a region of higher solute concentration from lower concentration to equalize the solute concentration on two sides. While in reverse osmosis, we apply external pressure upon concentrated solution to reverse the movement of solvent, i.e. from more concentrated to weaker dilute solution. This technique is used to purify the solution by making it free from impurities. Hence, both given statements are incorrect.

27. Consider the following statements with reference to osmosis :

(1) If the semipermeable membrane is placed between the solvent and solution, the solvent molecules will flow through the membrane from pure solvent to the solution.

(2) The process of flow of solvent is called as reverse osmosis.

Select the correct answer using the code given below :

- (a) Neither (1) nor (2) (b) Only (2)
(c) Only (1) (d) Both (1) and (2)

U.P. R.O./A.R.O. (Pre) 2023

Ans. (c)

If the semipermeable membrane is placed between the solvent and solution, the solvent molecules will flow through the membrane from pure solvent (lower concentration) to the solution (higher concentration). This process of flow of solvent is called osmosis, not reverse osmosis. Hence, only statement (1) is correct.

28. Which one of the following processes is NOT useful for purification of solid impurities?

- (a) Distillation (b) Sublimation
(c) Crystallization (d) All of the above

U.P. R.O./A.R.O. (Pre) (Re. Exam) 2016

Ans. (a)

Distillation is the action of purifying a liquid by a process of heating and cooling. Distillation is generally used for purification of liquid impurities while sublimation and crystallization are used for purification of solid impurities.

29. The technique of chromatography is used to –

- (a) Identify colour substances
(b) Determine the structure of substances
(c) Dry distillation of colouring substances
(d) Separate the substances from a mixture

42nd B.P.S.C. (Pre) 1997

Ans. (d)

Chromatography is a separation technique used to separate the different components in a liquid mixture. The separation is based on differential partitioning between the mobile and stationary phases. Chromatography may be preparative or analytical. The purpose of preparative chromatography is to separate the components of a mixture for more advanced use and thus it is a form of purification.

30. Process of solid camphor into camphor vapour is called:

- (a) Vaporization (b) Freezing
(c) Melting (d) Sublimation

44th B.P.S.C. (Pre) 2000

Ans. (d)

Solid camphor directly changes into camphor vapour and this process is called sublimation. The sublimation is the process in which some substances upon heating directly changes into vapour form and when cooled again directly turns into solid form.

31. A sample of any of the following substances disappears after sometime when exposed to air like camphor, naphthalene or dry ice. This phenomenon is called :

- (a) Sublimation (b) Evaporation
(c) Diffusion (d) Radiation

U.P.P.C.S. (Pre) 2019

Ans. (a)

Sublimation is the transition of a substance directly from the solid to the gas phase, without passing through the intermediate liquid phase. Substances like camphor, naphthalene or dry ice (solid carbon dioxide) display this phenomenon.

32. The conversion of a solid directly into gas is called as :

- (a) Sublimation (b) Condensation
(c) Evaporation (d) Boiling
(e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (a)

See the explanation of above question.

33. A mixture of sand and naphthalene can be separated by :

- (a) sublimation (b) distillation
(c) chromatography (d) fractional distillation
(d) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (a)

Naphthalene displays the phenomenon of sublimation, which is the transition of a substance directly from the solid to gas phase upon heating. Hence, a mixture of sand and naphthalene can be separated by sublimation.

34. Match List-I (oxidation number) with List-II (the element) and select the correct answer using the codes given below the lists :

List-I	List-II
A. 2	1. Oxidation number of Mn in MnO_2
B. 3	2. Oxidation number of S in H_2SO_4
C. 4	3. Oxidation number of Ca in CaO
D. 6	4. Oxidation number of Al in NaAlH_4

Code :

	A	B	C	D
(a)	3	4	1	2
(b)	4	3	1	2
(c)	3	4	2	1
(d)	4	3	2	1

I.A.S. (Pre) 2002

Ans. (a)

The oxidation number of any compound is always zero. This means that sum of the oxidation number of all the elements present in the compound is zero. By this rule, we can now find the oxidation number in given options here –

- (1) Suppose, the oxidation number of Mn in MnO_2 is x . MnO_2 contains two oxygen ions, each with an oxidation number (-2) .
 $\Rightarrow x + 2 \times (-2) = 0$
 $x - 4 = 0 \Rightarrow x = 4$
- (2) Suppose, the oxidation number of S in H_2SO_4 is x . H_2SO_4 contains two hydrogen ion, each with an oxidation number $(+1)$ and oxidation number of oxygen (-2) .
 $2(+1) + x + 4(-2) = 0$
 $\Rightarrow 2 + x - 8 = 0 \Rightarrow x = 6$
- (3) The oxidation number of Ca in CaO when oxygen has oxidation number (-2)
 $\Rightarrow x - 2 = 0 \Rightarrow x = 2$
- (4) Suppose, the oxidation number of Al in NaAlH_4 is x . NaAlH_4 contains one Na which has oxidation number $(+1)$ and four H which has oxidation number (-1) .
 $\Rightarrow (+1) + x + 4(-1) = 0$
 $\Rightarrow 1 + x - 4 = 0 \Rightarrow x = 3$

35. Which of the following slows down the reaction rate?

- (a) Catalytic promoter (b) Homogeneous catalyst
 (c) Heterogeneous catalyst (d) None of the above

69th B.P.S.C. (Pre) 2023

Ans. (d)

Catalytic promoter or accelerators are those substances used to increase the activity of catalysts positively. Reactions that are facilitated by catalysts can be divided into two major classes : homogeneous catalysis and heterogeneous catalysis.

A homogeneous catalyst is present in the same phase as the reactants. A heterogeneous catalyst is present in a different phase from the reactants. All of these facilitate reactions. Hence, option (d) is correct.

36. Which of the following are exothermic processes ?

- (i) Evaporation of water.
 (ii) Dilution of an acid (H_2SO_4).
 (iii) Reaction of water with quick lime.
 (iv) Sublimation of camphor (crystals).
 (a) (iii) and (iv) (b) (ii) and (iii)
 (c) (i) and (iv) (d) (i) and (ii)

70th B.P.S.C. (Pre) 2024

Ans. (b)

Exothermic reactions or processes release heat. When quicklime (CaO) reacts with water, it forms calcium hydroxide [$\text{Ca}(\text{OH})_2$] and release a large amount of heat. This reaction is highly exothermic. When a concentrated acid is diluted with water, heat is released. This is also an exothermic process, particularly for strong acids like sulphuric acid (H_2SO_4). Evaporation of water and sublimation of camphor are both endothermic reactions, meaning they absorb heat from their surroundings to proceed.

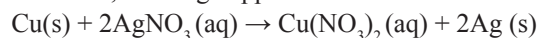
37. Which of the following pairs will give displacement reactions ?

- (a) AgNO_3 solution and copper metal
 (b) NaCl solution and copper metal
 (c) FeSO_4 solution and silver metal
 (d) MgCl_2 solution and aluminium metal

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Ans. (a)

A chemical reaction in which a more reactive metal displaces a less reactive metal is called displacement reaction. Copper is more reactive than silver, meaning it has a stronger tendency to lose electrons. When copper metal is placed in a silver nitrate solution, a displacement reaction occurs where copper replaces silver, forming copper nitrate and solid silver.



38. Example of corrosion is –

- (a) Black coating on Silver
 (b) Green coating on Copper
 (c) Brown coating on Iron
 (d) Above three
 (e) None of the above

Chhattisgarh P.C.S. (Pre) 2016

Ans. (d)

Gradual decay of metals by air or chemical substances is called corrosion. Rusting on iron i.e. brown coating on iron, green coating on copper and black coating on silver, all these are examples of corrosion.